Objectives

Features of Java.

Features of Java

- ✓ Simple
- ✓ Object-Oriented
- ✓ Portable
- ✓ Platform independent
- ✓ Secured
- ✓ Robust
- ✓ Architecture neutral
- ✓ Interpreted
- ✓ High Performance
- ✓ Multithreaded
- ✓ Distributed
- ✓ Dynamic

Features of Java

Simple

- ✓ Easy to learn, simple syntax, clean and easy to understand.
- ✓ Removed many complicated and rarely-used features, for example, explicit pointers, operator overloading, etc.
- ✓ Automatic Garbage Collection in Java removes unreferenced objects.

Features of Java

Object-oriented

- ✓ Object-oriented programming (OOP) is a computer programming model that organizes software design around data, or objects, rather than functions and logic.
- ✓ Java is an OOP language where everything is considered as an object.

Features of Java

Platform Independent

- ✓ A platform is the hardware or software environment in which a program runs.
- ✓ Java code is compiled by the compiler and converted into bytecode. This bytecode is a platform-independent code because it can be run on multiple platforms.
- ✓ Write Once and Run Anywhere(WORA)

Features of Java

Secured

- ✓ Java is best known for its security. With Java, we can develop virus-free systems. Java is secured because:
  - ✓ No explicit pointer
  - ✓ Java Programs run inside a virtual machine sandbox
  - ✓ Class loader
  - ✓ Bytecode Verifier
  - ✓ Security Manager

Features of Java

Robust

- ✓ Robust simply means strong. Java is robust because:
  - ✓ It uses strong memory management.
  - ✓ There is a lack of pointers that avoids security problems.
  - ✓ There is automatic garbage collection.
  - ✓ There is exception handling and the type checking mechanism in Java.

Features of Java

Architecture-neutral

- ✓ Java is architecture neutral because there are no implementation dependent features, for example, the size of primitive types is fixed.
- ✓ In C programming, int data type occupies 2 bytes of memory for 32-bit architecture and 4 bytes of memory for 64-bit architecture. However, it occupies 4 bytes of memory for both 32 and 64-bit architectures in Java.

Features of Java

Portable

Java is portable because it facilitates you to carry the Java bytecode to any platform. It doesn't require any implementation.

High-performance

Java is faster than other traditional interpreted programming languages because Java bytecode is "close" to native code.

Features of Java

Distributed

- ✓ Java is distributed because it facilitates users to create distributed applications in Java.
- ✓ RMI and EJB are used for creating distributed applications. This feature of Java makes us able to access files by calling the methods from any machine on the internet.

Features of Java

Multi-threaded

- ✓ A thread is like a separate program, executing concurrently.
- ✓ Java allows multi threading - which doesn't occupy memory for each thread.
- ✓ Threads shares a common memory area.
- ✓ Threads are important for multi-media, Web applications, etc.

Features of Java

Dynamic

- ✓ Java is a dynamic language. It supports dynamic loading of classes. It means classes are loaded on demand.
- ✓ It also supports functions from its native languages, i.e., C and C++.

Takeaways

- ✓ Features of Java

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition", Oracle 2018*